The Reading of Potential

Decompiling Binary Ring Topologies, Galois Core Invariants, and Three-Dimensional Matter in the NEXUS Compiler Space

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Executive Abstract and the Reverse-the-Gap Decompilation Principle

Conventional mathematical physics and theoretical computer science operate in a forward, accumulative direction: axiomatic primitives are posited, operations are applied sequentially, and terminal scalar values are evaluated as the primary ontology. Whenever a forward model encounters an underdetermined transition, an irreversible merge, or an apparent randomness in digit generation, traditional analysis attributes the incompleteness to the object itself. The research programme synthesized across the NEXUS corpus—spanning token-arena cohomology, Myhill–Nerode cut complexity, Bailey–Borwein–Plouffe (BBP) radix dynamics, and finite-field register topologies—establishes the exact inverse paradigm. The underlying transformation domain is complete at step zero. Every finite word of every length, every coordinatewise displacement, and every lawful transition between states already exists within a single, label-free graded substrate. Nothing is accumulated, constructed, or destroyed; structure is rendered exclusively by reading what is already present.

The operational method required to analyze a pre-compiled substrate is the reverse-the-gap decompilation process. Rather than building forward toward an object, a decompiler begins with the finished observation, subtracts the forward transport route, and binds the surviving residual to its structural owner in the domain. Under this principle, a gap is never a missing piece of reality; it is the signature of a reader's finite cut, projection fiber, or declared equivalence relation. While a collapsed value represents a settled boundary reading (0 or 1), potential—conceptualized as the uncollapsed . 5 between binary extremes—is the active space of admissible continuations held within a configuration fiber prior to scalar projection.

Ledger Classification	Operational Definition in the Decompilation Protocol	Canonical Instantiation Across the NEXUS Corpus
STANDS	Established theorem, algebraic identity, or empirical dataset supplied externally and cited as an anchor.	Burnside/Gauss necklace counts, Landauer erasure floor ($k_B T \ln 2$), Schwinger spin construction, Lucas Fermat prime splitting, Wootters–Fields mutually unbiased bases.

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Ledger Classification	Operational Definition in the Decompilation Protocol	Canonical Instantiation Across the NEXUS Corpus
RUNS	Computationally certified law where the formal route generated the invariant prior to or independently of data linkage.	Exhaustive 28/28 kernel checks in nexus_compiler_space.py, 47/47 Galois core checks in §133, 10,000 reversible SHA-256 round inversions, and atomic closure capacities $2,10,18,36,54,86,118$.
DEAD END	A hypothesis or forward formula that executed cleanly but failed empirical or algebraic validation, subsequently routed to its true structural owner.	Naive log-relation counts off the register chain ($n = 12,24$), order-independent maximal subgroup chains in G_{24}, and backward multi-round digest factorization in SHA-256.
OPEN	A sealed boundary question where the formal correspondence map or physical selector has not yet been written.	First-principles derivation of Yukawa mass couplings y_f, the atomic-versus-nuclear intruder direction selector, and forward constraint propagation across the 10-word SHA-256 front cone.
PULL	An interpretive bridge connecting an exact finite compiler invariant to a broader physical or ontological thesis.	Identifying the Baker's map stretch-and-fold axes with the dual wave of expanding compilation and collapsing observation, and equating Landauer erasure cost with physical anti-aliasing.

Holding both the expanding compiler view and the collapsing reader view simultaneously resolves long-standing dualities across discrete mathematics and physics. What appears from the outside as an inert, unmeasurable sphere of symbol counts is revealed from the inside as a non-abelian network of traversal routes whose reciprocal fiber sizes reconstruct the global surface without ever leaving the interior.

Decompiling the 2-Bit Ring: Non-Linear Folding, Step Gaps, and 8-Bit Runtime Reflection

The foundational dynamics of the compiler space emerge in full miniature within Ring 2, the 2-bit circular substrate documented in the two_bit_circle.xlsx and nested_rings artifacts. At bit length $n = 2$, the graded domain layer $S[2,2] = \{0,1\}^2$ contains exactly $2^2 = 4$ distinct words: 00 (assigned to the top-right quadrant, slices 1–2), 10 (bottom-right, slices 3–4), 01 (bottom-left, slices 5–6), and 11 (left-top, slices 7–8). When these four 2-bit states are arranged contiguously around a circle with radial thickness equal to their length (LEN $=$ 2), they partition the ring into $N = n \cdot 2^n = 2 \cdot 4 = 8$ discrete angular slices. Drawn clockwise from the top,

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the raw bit sequence around the perimeter reads 00100111; under the dual sheet reading—which traverses the circle in the reverse orientation with 0 and 1 inverted—the 8-slice sampled cycle is 00011011.

Tracing a sliding reader window of sample size $L \in \{2,3,4,5,6,7,8\}$ across the 8-slice cycle $s = (0,0,0,1,1,0,1,1)$ exposes four interconnected structural laws that govern the entire compiler space: non-linear value progression, two-dimensional value folding, deterministic step-gap doubling, and complete runtime reflection at the 8-bit boundary.

Non-Linear Value Traversal and Two-Dimensional Folding

When a reader of width $L = 2$ steps clockwise slice-by-slice around 00011011, the sequence of 2-bit words extracted is 00, 00, 01, 11, 10, 01, 11, 10, which evaluates in decimal to $0,0,1,3,2,1,3,2$. Rather than accumulating linearly ($0 \to 1 \to 2 \to 3$), the ring executes a folded Gray-code trajectory ($0 \to 1 \to 3 \to 2$) perforated by an initial echo ($0 \to 0$) and a repeating loop ($1 \to 3 \to 2 \to 1 \to 3 \to 2$). This non-linearity is not an irregularity; it is the direct consequence of reading a continuous circular boundary through a finite shifting aperture.

Let $w_i^{(L)} = \sum_{j=0}^{L-1} s_{(i+j) \bmod 8}\, 2^{L-1-j}$ denote the integer value of the clockwise window of length L starting at slice $i \in \{0, \ldots, 7\}$. The system exhibits an exact two-dimensional folding recurrence that links angular rotation (stepping $i \to i+1$ around the ring) with radial depth (expanding the reader width $L \to L+1$ across layers):

$$w_i^{(L+1)} = 2w_i^{(L)} + s_{(i+L) \bmod 8}$$

$$w_{(i+1) \bmod 8}^{(L)} = w_i^{(L+1)} - 2^L s_i = 2w_i^{(L)} - 2^L s_i + s_{(i+L) \bmod 8}$$

Because the first three slices of 00011011 are zeros ($s_0 = s_1 = s_2 = 0$), shifting the window angularly from slice 0 to slice 1 at width L is numerically identical to expanding the window radially from L to $L+1$ at slice 0: $w_0^{(L+1)} = w_1^{(L)}$ for every $L \geq 2$. Specifically, the value at row 1 of sample size L folds directly into row 0 of sample size $L+1$ across the entire hierarchy: $w_1^{(2)} = 0 = w_0^{(3)}$, $w_1^{(3)} = 1 = w_0^{(4)}$, $w_1^{(4)} = 3 = w_0^{(5)}$, $w_1^{(5)} = 6 = w_0^{(6)}$, $w_1^{(6)} = 13 = w_0^{(7)}$, and $w_1^{(7)} = 27 = w_0^{(8)}$. Furthermore, the sequence of initial values read radially across depths $L = 2 \ldots 8$ at slice 0—namely $(0,0,1,3,6,13,27)$—is the exact prefix of the angular doubling trajectory generated within the 8-bit ring itself ($27 \to 54 \to 108 \to 216$). In this geometry, radial compilation (increasing bit length L) and angular execution (rotating around the ring) are two projections of a single folding operation.

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Slice Index (i)	L=2 CW (Dec / Gap)	L=3 CW (Dec / Gap)	L=4 CW (Dec / Gap)	L=5 CW (Dec / Gap)	L=6 CW (Dec / Gap)	L=7 CW (Dec / Gap)	L=8 CW (Dec / Gap)	L=8 CW Hex / Nib	L=8 CCW Hex / Nib
0	00 (0/ 0)	000 (0/ +1)	0001 (1/ +2)	00011 (3/ +3)	000110 (6/ +7)	0001101 (13/ +14)	00011011 (27/ +27)	1B / 1B36	D8 / —
1	00 (0/ +1)	001 (1/ +2)	0011 (3/ +3)	00110 (6/ +7)	001101 (13/ +14)	0011011 (27/ +27)	00110110 (54/ +54)	36 / —	B1 / B163
2	01 (1/ +2)	011 (3/ +3)	0110 (6/ +7)	01101 (13/ +14)	011011 (27/ +27)	0110110 (54/ +54)	01101100 (108/ +108)	6C / 6CD8	63 / —
3	11 (3/ −1)	110 (6/ −1)	1101 (13/ −2)	11011 (27/ −5)	110110 (54/ −10)	1101100 (108/ −20)	11011000 (216/ −39)	D8 / —	C6 / C68D
4	10 (2/ −1)	101 (5/ −2)	1011 (11/ −5)	10110 (22/ −10)	101100 (44/ −20)	1011000 (88/ −39)	10110001 (177/ −78)	B1 / B163	8D / —
5	01 (1/ +2)	011 (3/ +3)	0110 (6/ +6)	01100 (12/ +12)	011000 (24/ +25)	0110001 (49/ +50)	01100011 (99/ +99)	63 / —	1B / 1B36

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Slice Index (i)	L=2 CW (Dec / Gap)	L=3 CW (Dec / Gap)	L=4 CW (Dec / Gap)	L=5 CW (Dec / Gap)	L=6 CW (Dec / Gap)	L=7 CW (Dec / Gap)	L=8 CW (Dec / Gap)	L=8 CW Hex / Nib	L=8 CCW Hex / Nib
6	11 (3/ −1)	110 (6/ −2)	1100 (12/ −4)	11000 (24/ −7)	110001 (49/ −14)	1100011 (99/ −29)	11000110 (198/ −57)	C6 / C68D	36 / —
7	10 (2/ −2)	100 (4/ −4)	1000 (8/ −7)	10001 (17/ −14)	100011 (35/ −29)	1000110 (70/ −57)	10001101 (141/ −114)	8D / —	6C / 6CD8

Consistent Step Gaps and the Frobenius Doubling Orbit

Analyzing the step differences $\Delta_i^{(L)} = w_{(i+1) \bmod 8}^{(L)} - w_i^{(L)}$ across the window hierarchy reveals why the gaps remain structurally consistent despite the non-linear state values. For any window size L, the displacement between adjacent readings is governed by the difference between the entering bit $s_{(i+L) \bmod 8}$ and the exiting bit s_i:

$$\Delta_i^{(L)} = w_i^{(L)} - \left(2^L s_i - s_{(i+L) \bmod 8}\right)$$

When the reader width reaches $L = 8$, the window encompasses all $N = 8$ slices of the ring. Because $s_{(i+8) \bmod 8} = s_i$ identically, the entering bit is the exiting bit, and the correction term collapses to $(2^8 - 1)s_i = 255s_i$. Consequently, at the 8-bit closure, the step gap obeys an exact piecewise linear law:

$$\Delta_i^{(8)} = \begin{cases} +w_i^{(8)} & \text{if } s_i = 0 \quad (\text{slices } 0,1,2,5: \Delta = +27, +54, +108, +99) \\ w_i^{(8)} - 255 & \text{if } s_i = 1 \quad (\text{slices } 3,4,6,7: \Delta = -39, -78, -57, -114) \end{cases}$$

Every positive gap equals the state value itself, and every negative gap doubles across consecutive 1-bits ($-39 \to -78$ across slices $3 \to 4$, and $-57 \to -114$ across slices $6 \to 7$) because $(2w_i^{(8)} - 255) - 255 = 2(w_i^{(8)} - 255)$. Modulo $2^8 - 1 = 255$, the transition is pure multiplication by 2: $w_{i+1}^{(8)} \equiv 2w_i^{(8)} \pmod{255}$. As established in *One Program, Infinite Compilers*, in a normal basis of the finite field $\mathbb{F}_{2^8}$, a one-bit cyclic rotation of an 8-bit coordinate word is identically the Frobenius squaring map $x \mapsto x^2$. Thus, the 8-slice 2-bit ring is simultaneously a spatial concatenation of the four 2-bit words and a single 8-element Frobenius orbit in $\mathbb{F}_{2^8}$ generated by $27 = 3^3$ ($27 \equiv 27 \cdot 2^8 \pmod{255}$), where word rotation and algebraic squaring are one transformation read twice.

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8-Bit Runtime Reflection and Boolean Gate Unification

The central phenomenon in the 2-bit ring dataset occurs at $L = 8$, annotated in the laboratory ledger as pure reflection in the hexadecimal cross-pairs. This reflection operates simultaneously on two levels: within the 8-slice ring's bidirectional geometry and across the global NEXUS function-space hierarchy.

Within the 8-slice ring, the Counter-Clockwise (CCW) reader begins at slice offset 3 (where the first 11 lands) and steps backward. At window length $L = 8$, every clockwise word $w_i^{(8)}$ is a cyclic shift of 00011011, and every counter-clockwise word $\widetilde{w}_i^{(8)}$ is a cyclic shift of the reversed string 11011000. Because the underlying 8-bit word 00011011 (27, 0x1B) is palindromically symmetric under a 3-step shift—that is, reversing the bits of 00011011 yields 11011000 (216, 0xD8), which is already the clockwise state at slice 3 ($w_3^{(8)} =$ '11011000')—the set of 8 clockwise bytes and the set of 8 counter-clockwise bytes are identical as sets, offset by exactly 3 steps:

$$\widetilde{w}_i^{(8)} = w_{(i+3) \bmod 8}^{(8)} \quad \text{and} \quad \mathrm{Rev}_8\left(w_i^{(8)}\right) = \widetilde{w}_{(8-i) \bmod 8}^{(8)} = w_{(11-i) \bmod 8}^{(8)}$$

In hexadecimal, the clockwise cycle 1B, 36, 6C, D8, B1, 63, C6, 8D and the counter-clockwise cycle D8, B1, 63, C6, 8D, 1B, 36, 6C form four exact bit-reversal reflection pairs: 1B ↔ D8, 36 ↔ 6C, B1 ↔ 8D, and 63 ↔ C6. When consecutive bytes are paired into 16-bit hexadecimal words (Hex nib), the four clockwise pairs (1B36 at slice 0, 6CD8 at slice 2, B163 at slice 4, C68D at slice 6) reappear intact in the counter-clockwise reading at odd slice positions (B163 at slice 1, C68D at slice 3, 1B36 at slice 5, 6CD8 at slice 7). Simultaneously, the seven Boolean logic columns (Not, And, Nand, Or, Nor, Xor, Xnor), which exhibit position-dependent variations at partial window widths $L = 2 \ldots 7$, collapse at $L = 8$ into complete rotational uniformity across all 8 rows (And = 0, Nand = 1, Or = 0, Nor = 1, Xor = 0, Xnor = 1), while row 3 of the Not column outputs 00100111—recovering the exact clockwise top-drawn circle from the inverted sheet cycle.

At the global compiler-space level, nexus_compiler_space.py proves why 8 bits is the universal threshold of runtime reflection for a 2-bit system. A total transformation $T: S[b, n] \to S[b, m]$ requires a truth table of length $m \cdot b^n$ digits in base b, yielding $b^{m \cdot b^n}$ distinct functions. For a 2-bit endomorphism ($b = 2, n = 2, m = 2$), the truth table length is $2 \cdot 2^2 = 8$ bits, and the number of possible transformations is $2^8 = 256$. Thus, the set of all 256 operators $T[2{,}2 \to 2]$ is in canonical bijection with the set of all 256 words in the 8-bit layer $S[2{,}8]$. At $L = 8$, the ring's state space and the 2-bit operator space become the exact same object: reading an 8-bit window on the 2-bit ring is literally reading the compiled truth table of a 2-bit transformation.

Multi-Base Sampling, Reader Steps, and the Three Bases of 64

In the 2-bit ring table, each binary window is evaluated across three non-binary numeral columns: Decimal (Dec, base 10), Hexadecimal (Hex, base 16), and Octal (Oct, base 8). Alongside this empirical table, the theoretical audit in *One Program, Infinite Compilers* investigates how base selection operates across the 64-bit register stack. Decompiling the relationship between a binary substrate and its numeral representations demonstrates that grouping bits into a base adds zero distinctions to the underlying field; instead, a vocabulary is strictly the width of the reader's step.

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For a register of length LEN, a reader with block width w traverses the word in steps $=$ LEN$/w$, satisfying the invariant steps × width $=$ LEN. In an 8-bit register, shifting in destination digits of width $w \in \{1,2,4\}$ bits (corresponding to Base $2^1 = 2$, Base $2^2 = 4$, and Base $2^4 = 16$) routes every one of the 256 starting states to every destination in exactly 8, 4, and 2 steps, respectively. The destinations reached, the collision counts at each step, and the 256 trajectories converging into each address are identical across all three readers; ungrouping the blocks returns the exact binary route without loss.

Candidate Base (bbb)	**Block Width (w=log 2bw = \log_2 bw=log2b)**	**Mean Digits Altered per 1-Bit Flip (64-Bit Word)**	**Max Digits Altered per 1-Bit Flip**	**Divides All Registers 4…644 \dots 644…64?**	**Structural Classification in Compiler Space**
Base 2 (Binary)	1 bit	1.00	1	Yes ($1 \mid 2^k$)	Pure whole-block reading (fundamental bit step).
Base 4 (Quaternary)	2 bits	1.00	1	Yes ($2 \mid 2^k$ for $k \geq 1$)	Pure whole-block reading (2-bit ring step).
Base 8 (Octal)	3 bits	1.00	1	No ($3 \nmid 4,8,16,32,64$)	Local whole-block reading; misaligned with 2^k register boundaries.
Base 16 (Hexadecimal)	4 bits	1.00	1	Yes ($4 \mid 2^k$ for $k \geq 2$)	Pure whole-block reading; half-turn of the byte and closure of $s^8 = 1/16$.

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Candidate Base (bbb)	Block Width (w=log 2bw = \log_2 bw=log2b)	Mean Digits Altered per 1-Bit Flip (64-Bit Word)	Max Digits Altered per 1-Bit Flip	Divides All Registers 4…644 \dots 644…64?	Structural Classification in Compiler Space
Base 256 (Byte)	8 bits	1.00	1	Divides 8,16,32,64	Pure whole-block reading; full 2-bit operator table word.
Base 3 (Ternary)	≈ 1.585 bits	14.97	35	No (non-power of 2)	Global conversion; native to repeat-free frames (e.g., SonicDMX).
Base 10 (Decimal)	≈ 3.322 bits	9.48	20	No (introduces prime 5)	Global conversion; couples single-bit flips across up to 20 digits.
Base 22	≈ 4.459 bits	7.66	15	No (introduces prime 11)	Global conversion; wastes 10 of 32 5-bit slots, route length 10 never closes.

This block-divisibility criterion explains the sharp dichotomy verified in 4,000 seeded 64-bit single-bit-flip trials. When a single bit is flipped in a 64-bit word, whole-block power-of-two bases (2,4,8,16,256) alter exactly 1.00 digit every time, because the flipped bit is confined to a single disjoint block. Conversely, bases that are not powers of two—such as Base 3, Base 10 (Dec), or Base 22—are *conversions* rather than readings: flipping a single bit cascades across an average of 14.97 digits in Base 3 (maximum 35), 9.48 digits in Base 10 (maximum 20), and 7.66 digits in Base 22 (maximum 15), because every converted digit depends globally on

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the entire word. Furthermore, among power-of-two readings, only the three sub-byte widths $w \in \{1,2,4\}$ (Bases 2, 4, and 16) tile every register in the chain from 4 to 64 bits. Octal ($w = 3$, Base 8) is a genuine local reading (changing 1 digit per bit flip), but because 64 is a power of 2, $3 \nmid 64$; an octal reader leaves an unaligned remainder at the boundary of 4,8,16,32, and 64-bit registers.

The interplay between Base 8 (Octal) and Base 16 (Hexadecimal) carries a deeper algebraic consequence when examined through the mirror-box polynomial in Latest_published_papers.docx and the Baker's map in *One Program, Infinite Compilers*. When the octal closure polynomial $C(y) = y(y^2 + 2)(y^2 - 4y + 2)(y^2 + 2y + 2)$ is evaluated at $y = +1$, the visible odd prime factors are 3 and 5—precisely the Fermat primes of Base 16 ($16 - 1 = 15 = 3 \times 5$). When reflected across the mirror to $y = -1$, the factor $y^2 - 4y + 2$ evaluates to 7, exposing the primes 3 and 7, where $7 = 8 - 1$ is the unshared Mersenne prime of the octal/hexagonal seed. Simultaneously, comparing one Base-16 step with four Base-2 steps on the invertible Baker's map $(x, y) \mapsto (\{bx\}, (y + \lfloor bx \rfloor)/b)$ proves that while the expanding shape coordinate x is identical after four binary steps and one hex step ($2{,}000/2{,}000$ exact rational trials), the collapsing history coordinate y stores the 4-bit hex digit bit-reversed ($2{,}000/2{,}000$ trials), agreeing only on the four palindromic nibbles (0000, 0110, 1001, 1111). Naming a 4-bit block and reading it in one hexadecimal step preserves forward route order inside the nibble; reading it via narrow 1-bit steps records history as the reflection of the route.

The Graded Compiler Kernel, Instruction-Set Closure, and the 64-Bit Register Stack

In nexus_compiler_space.py, the entire architecture of finite words, differences, transformations, and higher-order compilers is unified into a single executable domain schema without external assumptions. For any integer base $b \geq 2$ and length $n \geq 0$, the state layer is $S[b, n] = \{0,1, \dots, b - 1\}^n$ of cardinality b^n.

The Difference Kernel and Recursive Compiler-Table Growth

The coordinatewise difference operator $\Delta: S[b, n] \times S[b, n] \to S[b, n]$ defined by $\Delta(x, y) = (y_i - x_i) \bmod b$ satisfies three foundational properties verified across 22 base/length configurations ($b = 2, n = 0 \dots 8$; $b = 3, n = 0 \dots 5$; $b = 4, n = 0 \dots 4$) and $89{,}405$ recursive edge checks. First, by onto isomorphism, the image of Δ is the entire word layer $S[b, n]$ itself, establishing that displacements do not inhabit a separate mathematical space outside the state inventory. Second, by uniform preimage fibration, for every displacement pattern $\delta \in S[b, n]$, the fiber $\Delta^{-1}(\delta) = \{(x, y): y - x \equiv \delta \pmod b\}$ contains exactly b^n ordered pairs, because $y = (x + \delta) \bmod b$ is uniquely determined for each $x \in S[b, n]$. Third, by edge-split recursion, appending a new digit splits the difference cleanly at the boundary: $\Delta(x \| a, y \| c) = \Delta(x, y) \| ((c - a) \bmod b)$.

Because any total map $T: S[b, n] \to S[b, m]$ is completely specified by listing its b^n output words of length m in canonical lexicographic input order, every transformation is natively encoded as a single word of length $L_{\text{table}} = m \cdot b^n$ in the same base b. When $n = m = L$ (an endomorphism on layer L), its truth-table word has length $L' = L \cdot b^L$, which is itself a state in layer $S[b, L \cdot b^L]$. Iterating this wrapper encoding ($L_{d+1} = L_d \cdot 2^{L_d}$ for $b = 2$) generates an unbounded tower of finite compiler descriptions. Starting at depth 0 with $L_0 = 1$ bit, the operator table at depth 1 has length $L_1 = 1 \cdot 2^1 = 2$ bits ($2^2 = 4$ maps). At depth 2, $L_1 = 2$ bits expands to $L_2 = 2 \cdot 2^2 = 8$ bits ($2^8 = 256$ maps); at depth 3, $L_2 = 8$ bits expands to $L_3 = 8 \cdot 2^8 = 2{,}048$ bits (2^{2048} maps); at depth 4, $L_3 = 2{,}048$ bits expands to $L_4 = 2048 \cdot 2^{2048}$ bits (a 620-digit integer length); and at depth 5, $L_5 = L_4 \cdot 2^{L_4}$ continues symbolic strict growth since $L \cdot 2^L > L$ for all $L \geq 1$.

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Crucially, the sequence of endomorphism table lengths $L \cdot 2^L$ (2,8,24,64,160,384,896,2048, … ,49152 at $L = 12$) is identical to the total number of angular slices $n \cdot 2^n$ in the nested polar rings of nested_rings (3).ipynb. Thus, the slice circumference of Ring n in 3D space is literally the bit capacity required to write down the truth table of a single n-bit endomorphism.

Closure of the Instruction Set at Width Three

Examining the transformation landscapes across widths $n = 1,2,3,4$ demonstrates that the operational instruction set is not designed by external fiat; it is algebraically forced at the bottom and closes permanently at width three. At width 1 ($S[2,1] \to S[2,1]$, 4 maps), the 4 functions partition by image rank into 2 bijections of rank 2 (pass $x \mapsto x$ and flip $x \mapsto x \oplus 1$) and 2 constant merges of rank 1 (write-0 and write-1). Executing a 1-bit opcode on a 1-bit datum is identical to indexing the opcode's 2-bit truth table at the address supplied by the datum.

At width 2 ($S[2,2] \to S[2,2]$, 256 maps), enumerating all 256 maps yields image-rank distribution $\{1: 4, 2: 84, 3: 144, 4: 24\}$. Inherited 1-bit operations and wire swaps generate 36 two-step route histories over the 6-gate generating set $\{I, X_0, X_1$, SWAP, CNOT_{01}, $\text{CNOT}_{10}\}$, which collapse via relations such as $X_0; X_0 = I$ and SWAP; SWAP $= I$ into fewer distinct operators—proving that a composed operator table does not retain its route history. To span the full width-2 landscape, only two new primitives are forced: the asymmetric reversible mixer $(a, a \oplus b)$ (the Mold, which preserves full history with uniform fibers) and the single-pair merge (144 maps of rank 3, all AND-type). Combining the reversible mixer a ^ b and the irreversible merge a & b on two wires produces the half-adder, which loses exactly $w/2$ bits of order (every fiber is a swap orbit of size $2^{\text{differing bits}}$) while folding the carry onto a single wire costs 7.279 bits at $w = 8$.

At width 3 ($S[2,3] \to S[2,3]$, 40,320 reversible permutations), because every reversible 2-bit gate is affine over $\mathbb{F}_2$, embedding 2-bit gates into 3 lanes generates only the affine general linear group $\text{AGL}(3,2)$ of order 1,344 out of $8! = 40{,}320$ reversible states. The single forced primitive that closes the gap is the held carry—the 3-bit Toffoli gate $(a, b, c) \mapsto (a, b, c \oplus (a \wedge b))$—which completes all 40,320 permutations. Similarly, every de Bruijn ring of order $n \geq 3$ requires a non-linear feedback rule next = oldest $\oplus$ g(rest) where g embeds an AND-type merge inside the bijection. For all widths $n \geq 4$, gates of width $\leq n - 1$ generate the alternating group A_{2^n} (exactly half of all permutations), and borrowing a single auxiliary lane (returned unchanged) recovers the odd permutations. Thus, no new logical verb ever appears above width three; everything higher is pure length and routing.

The Register Stack from 64 Bits Down and Fermat Inversion Primes

Decompiling the register stack from 64 bits downward rather than building upward from 1 bit exposes why computer architecture is locked to powers of two. The 2^{64} words of length 64 partition by their least period d, which must divide 64. Denoting by $P_b(d) = \frac{1}{d}\sum_{k|d} \mu(k) b^{d/k}$ the number of primitive (aperiodic) rotation rings of length d in base b, the Burnside–Möbius identity

$$2^n = \sum_{d|n} d \cdot P_2(d)$$

holds exactly at every layer $n = 1 \ldots 64$. Because the divisors of 64 are exclusively the register widths 1,2,4,8,16,32,64, smaller registers are not historical stepping stones toward 64; they sit inside the 64-bit ring

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as the exact sets of periodic words that repeat. Specifically, the words of layer n left invariant by a half-turn rotation ($n/2$ steps) are precisely the duplicated words uu with $u \in S[2, n/2]$, which in a normal basis of $\mathbb{F}_{2^n}$ correspond to the subfield $\mathbb{F}_{2^{n/2}}$ fixed by the Frobenius involution $x \mapsto x^{2^{n/2}}$. Thus, each register layer inherits the entire layer below it and nothing else, with the inherited fraction decaying as $2^{-n/2}$ (from $1/2$ at $n = 2$ down to 2^{-32} at $n = 64$).

Register Width (LEN=d\text{ LEN} = dLEN=d)	**Total Words (2d2^d2d)**	**Inherited Periodic Words (=2d/2= 2^{d/2}=2 d/2)**	**New Primitive Rings P2(d)P_2(d)P2(d)**	**Total Angular Slices (d·2dd \cdot 2^dd·2 d)**	**New Fermat Factor (2d/2+12^{ d/2} + 12d/2+1) & Prime Factorization**
1	2	0	2	2	$2^0 + 1 = 2$ (Base distinction)
2	4	2	1	8	$2^1 + 1 = 3$ (First inversion prime)
4	16	4	3	64	$2^2 + 1 = 5$ (Nibble / Base-16 closure with $3 \times 5 = 15$)
8	256	16	30	2,048	$2^4 + 1 = 17$ (Byte / 2-bit operator reflection layer)

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Register Width (LEN=d\text{ LEN} = dLEN=d)	Total Words (2d2^d2d)	Inherited Periodic Words (=2d/2= 2^{d/2}=2 d/2)	New Primitive Rings P2(d)P_2(d)P2(d)	Total Angular Slices (d·2dd \cdot 2^dd·2 d)	New Fermat Factor (2d/2+12^{ d/2} + 12d/2+1) & Prime Factorization
16	65,536	256	4,080	1,048,576	$2^8 + 1 = 257$ (Half-word Fermat prime F_3)
32	4,294,967,29	65,536	134,215,680	1.374×10^{11}	$2^{16} + 1 = 65{,}537$ (Word Fermat prime F_4)
64	2^{64}	2^{32}	288,230,376,084,602,8	1.181×10^{21}	$2^{32} + 1 = 641 \times 6{,}700{,}417$ (F_5 composite split)

Furthermore, each doubling of the register width multiplies the multiplicative group order by a Fermat factor via $2^{2m} - 1 = (2^m - 1)(2^m + 1)$. The first factor $(2^m - 1)$ is the old layer; the second factor $(2^m + 1)$ introduces new primes, every single one of which satisfies $2^m \equiv -1 \pmod{p}$. Consequently, every prime native to the 64-bit register stack—namely $2^{64} - 1 = 3 \cdot 5 \cdot 17 \cdot 257 \cdot 641 \cdot 65537 \cdot 6700417$—enters the stack as an algebraic inversion (the half-period of an even loop). Primes whose multiplicative order of 2 contains an odd factor—such as 7 (order 3), 11 (order 10), 13 (order 12), 19 (order 18), 23 (order 11), 29 (order 28), and 31 (order 5)—are completely absent from the 64-bit register chain. Notably, width 3 (which brings in the Mersenne prime $2^3 - 1 = 7$ and closes the instruction set via the held carry) is not a register width at all.

The Join Theorem: Why Scale and Phase Returns Meet Only in Binary

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Section 133 of the research ledger (NEXUS_133_ask_the_field.md) seals and proves the universal theorem governing zero routes across all n-lane radix machines in base $b = p^{n/2}$. A lane machine evaluates linear combinations of the series $L_j = \sum_{k=0}^{\infty} b^{-k}/(nk + j)$ for $j = 1 \ldots n$, which expand over the n-th roots of unity $\zeta_n = e^{2\pi i/n}$ via the logarithmic wheel returns $L_m = -\log(1 - w_m)$, where $w_m = \zeta_n^m/\sqrt{p}$ and $\beta_m = \sqrt{p} - \zeta_n^m = \sqrt{p}(1 - w_m)$. Across all $m \in \{0, \ldots, n-1\}$, the product of the wheel returns reproduces the base-p register ring:

$$\prod_{m=0}^{n-1} \beta_m = \prod_{m=0}^{n-1} \left(\sqrt{p} - \zeta_n^m\right) = p^{n/2} - 1$$

A return β_m is prime-free (costs no prime factor in the ring of algebraic integers) if and only if it is an algebraic unit. Exactly two types of returns can be units. The first type consists of scale returns ($\zeta_n^m = \pm 1$), where the norm in $\mathbb{Q}(\sqrt{p})$ is $|N(\sqrt{p} \mp 1)| = p - 1$. This norm equals 1 if and only if $p = 2$; for every odd prime $p \geq 3$, the scale return pays an even factor $p - 1 \geq 2$, meaning binary sits inside every odd stack. The second type consists of phase returns (β_m a root of unity), where requiring $|\sqrt{p} - \zeta_n^m| = 1$ means two unit circles in $\mathbb{C}$ centered at 0 and $\sqrt{p}$ must intersect, which occurs if and only if $\sqrt{p} \leq 2 \Leftrightarrow p \in \{2,3\}$. Equivalently, $1 - w_m = \bar{w}_m$, which places the wheel at $\mathrm{Re}(w_m) = 1/2$ (the exact .5 potential midpoint). For $p = 2$, this occurs at $n = 8$ ($\theta = \pi/4$), where $\sqrt{2} - \zeta_8 = \zeta_8^{-1}$ and $w_1 = \frac{1+i}{2} = s$. For $p = 3$, it occurs at $n = 12$ ($\theta = \pi/6$), where $\sqrt{3} - \zeta_{12} = \zeta_{12}^{-1}$ and $w_1 = \frac{1}{2} + \frac{\sqrt{3}}{6} i$. For all primes $p \geq 5$, no phase return exists.

Therefore, binary ($p = 2$) is the unique prime in mathematics where both scale returns and phase returns are simultaneously prime-free. Their collision is the single modulus relation $a_0 + a_{n/2} = 2a_{n/8}$—stating that the scale pair $(\sqrt{2} - 1)(\sqrt{2} + 1) = 1 \Longrightarrow (1 - w_0)(1 - w_{n/2}) = 1/2$ and the phase pair $(1 - w_{n/8})(1 - w_{7n/8}) = |1 - s|^2 = 1/2$ each cost exactly one bit (1/2).

When testing off-chain layers such as $n = 12$ and $n = 24$, the naive hypothesis that the zero-route dimension $Z_p(n)$ equals the full $\mathbb{Q}$-relation space $\dim R$ among the logs L_m fails ($d_R = 2$ vs. measured $Z_2(12) = 1$, and $d_R = 7$ vs. measured $Z_2(24) = 5$). Decompiling this gap via Baker's theorem on linear forms in logarithms reveals that a rational machine annihilates an orbit tuple if and only if the relation is invariant under the twisted Galois action $\sigma_a : \beta_m \mapsto \pm\beta_{(am + \delta_a n/2) \bmod n}$. Galois descent yields the exact universal law:

$$Z_p(n) = \dim R_0, \qquad R_0 = \bigcap_{\sigma \in \mathrm{Gal}(\mathbb{Q}(\zeta_n, \sqrt{p})/\mathbb{Q})} \pi_\sigma(R)$$

At $n = 12$, the extra argument relation $2b_1 - b_3 - 4b_5 = 0$ is mapped by the Galois automorphism σ_7 to $6(b_1 - b_5) \neq 0$; thus it is visible to the complex returns at one embedding but invisible to any rational machine. On the binary register chain $n = 2^e$ ($e \geq 3$), the Fermat numbers $F_k = 2^{2^k} + 1$ are pairwise coprime, $\mathrm{ord}_{2^e}(3) = \mathrm{ord}_{2^e}(5) = 2^{e-2}$, and by Lucas's theorem every prime factor $q \mid F_k$ satisfies $q \equiv 1 \pmod{2^{k+2}}$, splitting completely in $\mathbb{Q}(\zeta_{2^e})$ so that every conjugate return owns a distinct prime ideal. Unique prime factorization kills all exponents on prime-bearing orbits, proving that $Z_2(2^e) = 1$ at every layer of the binary register stack, spanned exclusively by the pullback $\pi^* Z_2(8)$ of BBP's zero route.

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Modeling the Rings as Matter in Three-Dimensional Space

Modeling the nested rings as matter in three-dimensional space connects the polar slice geometry of Rings 1 through 12 (nested_rings (3).ipynb) directly to the spherical abelianization, Schwinger spin surfaces, and relativistic Klein–Gordon persistence established across*One Program, Infinite Compilers*, *The Inverted Mold*, and the *Running Audit*.

The Three Spatial Coordinates of the Compiler Sphere

To embed the abstract graded domain $S[2,n]$ into three-dimensional physical space, each structural operation of the kernel maps to an orthogonal geometric degree of freedom. The first coordinate is radial depth ($r = \text{LEN} = n$), which forms the nested ring stack and scale axis. As established in the polar visualizations of Rings 1–2, Rings 1–10, and Ring 12, each word length n forms a concentric shell of radial thickness n with zero radial gaps between successive layers. At layer n, the shell contains 2^n words partitioned into $n \cdot 2^n$ angular slices (2 slices at $n = 1$, 8 at $n = 2$, 24 at $n = 3$, 64 at $n = 4$, 160 at $n = 5$, 384 at $n = 6$, 896 at $n = 7$, 2,048 at $n = 8$, 4,608 at $n = 9$, 10,240 at $n = 10$, 22,528 at $n = 11$, and 49,152 at $n = 12$). Because a low-width k-bit transformation lifts into an n-bit state ($k \leq n$) by acting on the exposed k-bit prefix while preserving all 2^{n-k} hidden suffix contexts intact (lift_prefix and projection_fibers verified across 78 (n, k) pairs in nexus_compiler_space.py), outer radial shells encapsulate inner control laws without interference.

The second coordinate is the azimuthal angle (θ), which governs the internal route fibers and Frobenius phase traversal. Stepping around a ring at fixed radius n traverses the ordered history—the non-commutative sequence of arrivals, shifts, and gate compositions. In the interior of the sphere, words over k primitive generators form the free monoid of size k^n, where order is fully retained. As proven in fold.py via self-avoiding walks on the square lattice ($n = 5 \ldots 10$), the unordered occupied shape does not determine admissible future continuations (100% of multi-route shapes differ in future continuations at $n \leq 8$, and 97.31%–98.01% at $n = 9,10$); the minimal sufficient state requires the shape plus where the transformation currently stands. Similarly, in seam.py, analyzing /bin/ls ($142{,}312$ bytes $= 1{,}138{,}496$ bits) proves that the unordered symbol histogram fits in $2{,}690$ bits (0.24% of the file), whereas 99.76% of the program's address—and all $346{,}108$ bits of zlib-compressible structure—resides in the azimuthal ordering fiber, vanishing completely when shuffled within the same surface histogram.

The third coordinate is polar latitude (z), which defines the bit-sphere abelianization and weight projection. Mapping each binary digit $\{0,1\} \mapsto \{-1,+1\}$ embeds $S[2,n]$ onto the surface of an n-dimensional sphere of radius $\sqrt{n}$. Applying the order-forgetting projection (SURFACE in nexus_compiler_space.py and §99.4 of the *Running Audit*) quotients the free monoid by its commutator—forming the universal commutative quotient (abelianization)—which collapses the k^n internal routes onto $\binom{n+k-1}{k-1}$ surface points. For binary ($k = 2$), the surface consists of $n + 1$ discrete latitude bands indexed by Hamming weight $w \in \{0,1, \ldots, n\}$, where each latitude w holds a fiber of exactly $\binom{n}{w}$ ordered routes summing to $\sum_{w=0}^{n}\binom{n}{w} = 2^n$. Remarkably, a reader confined to the interior can measure the exterior surface area without ever touching a surface point: sampling random internal routes x, recording the reciprocal of their fiber size $1/|\text{fiber}(x)|$, and multiplying the expectation by k^n yields the exact Horvitz–Thompson identity $\sum_{x \in S[k,n]} \frac{1}{|\text{fiber}(x)|} = \binom{n+k-1}{k-1}$ (verified to $9.0/9$ at $k = 2, n = 8$; $28.0/28$ at $k = 3, n = 6$; $165.0/165$ at $k = 4, n = 8$; and $55.0/55$ at $k = 3, n = 9$).

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Light at the Poles, Rest Mass at the Equator, and the Wrapper Skin

The latitude fibration of the bit-sphere provides a direct, parameter-free geometric distinction between massless radiation at the polar caps and massive matter at the equatorial band. At the North and South poles ($w = 0$, 00…0, and $w = n$, 11…1), the fiber size is $\binom{n}{0} = \binom{n}{n} = 1$. There is only one ordering, zero internal permutation entropy, and no internal clock. In the relativistic 1-bit walker with only pass and flip, polar states never flip; they propagate strictly along the null cone $ds^2 = c^2dt^2 - dx^2 = 0$ where proper time vanishes ($d\tau = 0$). Because a massless excitation can never be brought to rest, it never encounters a homogeneous zero-gradient fixed point ($\nabla\phi = 0$) and requires no internal mass clock.

At the equator of an even layer $2m$ ($w = m$, equal numbers of 0s and 1s), the fiber reaches its maximum binomial multiplicity $\binom{2m}{m} \sim \frac{2^{2m}}{\sqrt{\pi m}}$, where π emerges inward as the asymptotic share of the field at its equator. Every equatorial ring contains at least one rotation from which it reads as a properly nested Dyck parentheses word ($(= 1,) = 0$), with Catalan$(m) = \frac{1}{m+1}\binom{2m}{m}$ closed wrapper objects at the band just above the equator (verified for $m = 0 \ldots 8$ in nexus_compiler_space.py). In the wrapper algebra (wrap $A \mapsto (A)$ and join $(A, B) \mapsto AB$), wrapping compiles an entire layer into a single closed skin one layer up, while joining computes horizontally. In physical space, an equatorial state has zero net spatial drift ($p = 0, \nabla\phi = 0$): to satisfy the C1 continuation constraint (which forbids null transitions or frozen states when spatial gradients vanish), the excitation must oscillate internally between its two readings (flip back and forth across the diameter). In the Klein–Gordon equation $(\Box + m^2)\phi = 0$ at $k = 0$, this internal equatorial turnaround is the Compton/Zitterbewegung frequency $\omega = mc^2/\hbar$ (7.76×10^{20} rad/s for the electron), while in the discrete edge operator $J_{ij}(t+1) = \alpha J_{ij}(t) - \beta(\rho_j - \rho_i) - \mu^2(\rho_i + \rho_j)/2$, the mass term $-\mu^2(\rho_i + \rho_j)/2$ breaks shift symmetry $\phi \to \phi + K$ by reading state amplitude rather than spatial difference. Matter ($x = x$) is therefore a persistent equatorial stalemate—light held back by a fold—whose internal turnaround ticks as rest mass.

Exact Decompilation of 3D Angular Momentum and the Periodic Table of Elements

In earlier stages of the audit (§99.5), the capacities of the atomic electron shells were recovered from three inputs: the Madelung degree grading $d = n + \ell$ on the 2-generator surface, the domain admissibility inequality $n_r = n - \ell - 1 \geq 0$ (which routes to the domain law $\ell < n$: *no part is ever the whole*), and the subshell orientation count $2(2\ell + 1)$, which had to be supplied externally. In *One Program, Infinite Compilers* (§6), the $2\ell + 1$ factor is decompiled directly from the doubled ring surface, eliminating the last external input.

Because a relational difference requires two positions (Co′, the two-frame requirement), a subshell part of radial size ℓ is read across a two-position gap as a doubled binary word of length 2ℓ bits. Projecting this 2ℓ-bit ring onto its order-free commutative surface leaves only the number of 1s, which takes values in $\{0,1, \ldots, 2\ell\}$—producing exactly $2\ell + 1$ discrete latitudes (verified for $\ell \leq 6$). This order-free surface is not merely a combinatorial match: constructing the collective spin operators S_x, S_y, S_z from the 2ℓ two-state bits (Schwinger's bosonic/qubit spin representation) proves that the total angular momentum squared satisfies

$$S^2 = S_x^2 + S_y^2 + S_z^2 = \ell(\ell + 1)I$$

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on every state of the symmetric order-free layer in exact rational arithmetic (verified for $\ell = 1,2,3,4$). Combining the 1-bit pass/flip reading factor (2, corresponding to Wootters–Fields $d = 2$ where a reading equals a single move) with the $2\ell + 1$ surface latitudes of the 2ℓ-bit ring and summing over all admissible parts $\ell < n$ yields the exact principal shell capacity:

$$\text{Capacity}(n) = \sum_{\ell=0}^{n-1} 2\,(2\ell + 1) = 2n^2 \Longrightarrow 2,8,18,32,50,72,98$$

When traversed along the 2-generator degree slices $d = n + \ell$ ($d = 1,2,\ldots,10$), the slice capacity is given by the exact closed form $C(d) = 2\lfloor (d+1)/2 \rfloor^2$, generating the paired period lengths $2,2,8,8,18,18,32,32,50,50$ and cutting at each new s-orbital to produce the exact noble-gas atomic closures $2,10,18,36,54,86,118$ (seal 478d1baf).

Madelung Degree (d=n+ℓd = n + \elld= n+ℓ)	**Admissible Subshells (nr=n−ℓ−1 ≥on_r = n - \ell - 1 \geonr =n−ℓ−1≥o)**	**Slice Capacity 2⌊(d+1)/2⌋22\lfloor(d+1)/2\rfloor^22⌊(d+1)/2⌋2**	**Cumulative Atomic Closure (Cut at New sss)**	**Nodeless Orbital (ℓ=n−1,nr =o\ell = n - 1, n_r = oℓ=n−1,nr =o)**	**Missing Predecessor at d−1d - 1d−1 (nr=−1n_r = -1nr =−1)**
1	$1s$	2	**2** (He)	$1s$ ($n = 1, \ell = 0$)	$(0,0) = 0s$ (Inadmissible)
2	$2s$	2	—	None (d even)	—
3	$2p, 3s$	8	**10** (Ne)	$2p$ ($n = 2, \ell = 1$)	$(1,1) = 1p$ (Inadmissible)
4	$3p, 4s$	8	**18** (Ar)	None (d even)	—

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Madelung Degree (d=n+ℓd = n + \elld= n+ℓ)	**Admissible Subshells (nr=n−ℓ−1 ≥on_r = n - \ell - 1 \geonr =n−ℓ−1≥o)**	**Slice Capacity 2⌊(d+1)/2⌋22\lfloor(d+1)/2\rfloor^22⌊(d+1)/2⌋2**	**Cumulative Atomic Closure (Cut at New sss)**	**Nodeless Orbital (ℓ=n−1,nr =o\ell = n - 1, n_r = oℓ=n−1,nr =o)**	**Missing Predecessor at d−1d - 1d−1 (nr=−1n_r = -1nr =−1)**
5	$3d, 4p, 5s$	18	**36** (Kr)	$3d$ ($n = 3, \ell = 2$)	$(2,2) = 2d$ (Inadmissible)
6	$4d, 5p, 6s$	18	**54** (Xe)	None (d even)	—
7	$4f, 5d, 6p, 7s$	32	**86** (Rn)	$4f$ ($n = 4, \ell = 3$)	$(3,3) = 3f$ (Inadmissible)
8	$5f, 6d, 7p, 8s$	32	**118** (Og)	None (d even)	—

The period doubling of the periodic table is forced by the floor function in $2\lfloor(d+1)/2\rfloor^2$, which in turn arises from a structural boundary obstruction (no_way_home.py): a nodeless orbital ($\ell = n - 1 \Longrightarrow n_r = 0$) has degree $d = n + (n-1) = 2n - 1$ and can therefore enter only at odd degree slices ($1s$ at $d = 1$, $2p$ at $d = 3$, $3d$ at $d = 5$, $4f$ at $d = 7$, $5g$ at $d = 9$). Its hypothetical same-ℓ predecessor one degree lower at $d - 1 = 2n - 2$ would have quantum numbers $(n-1, n-1)$ ($0s, 1p, 2d, 3f, 4g$), which requires $n_r = -1 < 0$—violating the domain law that a part cannot equal the whole. Because that predecessor cannot exist, every even slice $d = 2k$ adds zero new angular momentum types and must repeat the exact capacity of the odd slice $d = 2k - 1$ below it. In the nuclear shell table (nuclear_unfold.py, e2plus_independent.py), the same boundary piece—the partnerless top-j orbital of each harmonic oscillator shell (2,8,20,40,70,112)—is advanced rather than delayed by spin-orbit coupling, shifting the closures to the magic numbers 28,50,82,126,184 (verified across 954 ENSDF first-2^+ records and 675 even-even nuclei at $p = 2.11 \times 10^{-6}$ on the neutron axis and 5.29×10^{-6} on the proton axis).

Cut-Defined Existence, Two Quotients of One Machine, and the Cohomological Tower

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A central ontological conclusion of the corpus—established across *Two Quotients of One Machine* and *The Ceiling and the Descent*—is that the full, uncut domain has totality but zero internal continuation debt, and therefore cannot be a localized object relative to itself.

Cut-Defined Existence vs. Totality

Consider the n-bit binary adder reading input bit-pairs across coordinate positions $\{0,1,\dots,n-1\}$. When the read set is complete ($S=\{0,\dots,n-1\}$), the Myhill–Nerode continuation quotient is trivial: $F(S)=1$. Because no unread future positions remain that could make any past difference matter, the completed read collapses to a single class. Conversely, introducing a *cut* partitions the domain into a resolved past and an unresolved future ($\mathcal{D}=\mathcal{D}_{\text{past}}\cup\mathcal{D}_{\text{future}}$), creating a live boundary across which distinct past histories remain distinguishable if and only if they induce different future continuations.

For a cut-defined quotient Φ_S:Histories → Continuation Classes to constitute a persistent, autonomous entity rather than an arbitrary external summary, it must satisfy *local congruence closure*:

$$\Phi_S(\alpha)=\Phi_S(\beta)\Longrightarrow\Phi_{S\cup\{p\}}(\alpha[p\mapsto u])=\Phi_{S\cup\{p\}}(\beta[p\mapsto u])$$

When this congruence holds, the effective state updates locally via $(S,\Phi_S)\overset{(p,u)}{\to}(S\cup\{p\},\Phi_{S\cup\{p\}})$ without ever reopening the hidden fine history. In the $K=2$ witness field of order 24 (rings.py and rings2.py in §99.6), the subgroup lattice contains 26 subgroups and 36 maximal chains of cuts from the whole field down to a single point (12 short chains of length 3 with indices [4,3,2], and 24 long chains of length 4 with indices [2,3,2,2]). At every step, kept × leftover $=24$. Crucially, while 6 normal subgroups exist (the single folds of quotient sizes 24,12,6,3,2,1), the order-2 central subgroup and the order-4 normal subgroup inside A_4 do not nest. Consequently, zero maximal chains of cuts are order-independent ($0/12$ at length 3, $0/24$ at length 4), and 108 of the 276 proper subgroup pairs fail to commute. Order-independent cuts can reach the normal folds and no further; resolving an address past the fold level unconditionally requires the reader to supply a non-commuting measurement order.

Two Distinct Quotients of Binary Addition and Reader vs. Dynamical Depth

As synthesized in the twenty-paper architectural reviews and proven in *Two Quotients of One Machine*, a single operational machine supports multiple non-interchangeable quotients that must not be conflated into one scalar. The first is the behavioral address quotient (Myhill–Nerode cut complexity $F(S)$), which identifies partial assignments $\alpha\sim_{\Delta,\pi}\beta$ when no future input continuation distinguishes their readout. For an arbitrary subset $S\subset\{0,\dots,n-1\}$ of read bit positions, the minimal state count factors as an exact product over contiguous read runs $F(S)=\prod f(R)$, and for $n\geq 3$ the LSB-first reading order is the unique minimax schedule (with $n=2$ admitting the exact two-order tie). The second is the homological relation residual ($b_1^{\text{disj}}(n)$), which counts unclosed 1-cycles modulo the span of declared local 2-cells (disjoint-support commuting squares). For the n-bit binary adder, the disjoint-square locality cohomology has exact rank

$$b_1^{\text{disj}}(n)=\binom{n}{2}+(n-1)=\frac{(n-1)(n+2)}{2},\qquad \dim(B_A/B_D)=\binom{n}{2},\qquad \dim M_{ij}=n-j$$

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Both invariants collapse exponential state labels (2^{2n}) down to polynomial carry-seam statistics, yet they measure orthogonal structural features: $F(S)$ measures the active memory burden across a cut boundary, whereas $b_1^{\mathrm{disj}}(n)$ measures undeclared non-local relation directions across the carry chain.

Furthermore, examining how base change samples carry trajectories ($c_q^{(k)} = c_{qk}^{(1)}$, verified across $1{,}310{,}720$ trajectories at $N = 12$ with 0 mismatches, such as $x =$ 'oxAAA' alternating $(0,0,1,0,1,\dots)$ at odd rungs $k = 1,3$ and running flat $(0,1,1,1,\dots)$ at even rungs $k = 2,4$) establishes the fundamental separation between reader depth and dynamical depth. In the adder, the transfer maps have the closed form $c \mapsto \lfloor (c + A)/b^g \rfloor$, which saturates under composition. Refinement to a finer radix exposes finer provenance (reader depth: which fine word produced the transfer) without adding new operational capabilities (dynamical depth $= 0$). True dynamical depth arises only when refinement enlarges the compositional semigroup itself—as occurs when passing from width 1 to width 2 (CNOT/AND) and width 3 (Toffoli), or when lifting linear evolution into quadratic interaction on a deficiency module (Deficiency Is a Module).

The Cohomological Tower and Persistence Over Declared Relations

In *The Ceiling and the Descent*, the programme corrects the assumption that resolution and relation are two separate axes, proving instead that there is one logical operation—declare certain differences inessential, take the quotient, and measure the surviving residual—applied recursively across successive relational orders. At Order 0 (states, C^0), configurations x, x' are identified under a symmetry group G or reader partition, leaving distinguishable state orbits. At Order 1 (transitions, C_G^1), invariant 1-cochains ω are identified modulo exact endpoint potentials ($\omega \sim \omega + d_0 f$), leaving the transition residual of dimension $\varepsilon = \dim(C_G^1/\mathrm{im} d_0)$. At Order 2 (routes, C_G^2), pairs of paths are declared equivalent via a 2-cell predicate $\mathcal{K}$ (such as disjoint-support squares $\mathcal{K}_{\mathrm{disj}}$ vs. all 4-cycles $\mathcal{K}_4$), leaving the route residual $\varepsilon_{\mathrm{harm}} = \dim(\ker \bar{d}_1)$. At Order 3 (relations among relations, C_G^3), pairs of 2-cell identifications are declared compatible via commuting 3-cubes (generated by three pairwise vertex-disjoint token moves A, B, C with alternating face boundary $\partial[A,B,C] = [B,C]_{A=1} - [B,C]_{A=0} - [A,C]_{B=1} + [A,C]_{B=0} + [A,B]_{C=1} - [A,B]_{C=0}$ satisfying $d_2 d_1 = 0$ identically), leaving the second-order residual $\varepsilon_{\mathrm{harm}}^2 = \dim C_G^2 - \mathrm{rank}(d_1|_{C_G^1}) - \mathrm{rank}(d_2|_{C_G^2})$.

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Token Arena Substrate (KK Tokens)	dim CG1\dim C^1_GdimCG1	dim CG2\dim C^2_GdimCG2	Commuting Cubes / dim Z2\dim Z_2dim Z2	rank (d1)\operatorname{rank}(d_1)rank(d1)	rank (d2)\operatorname{rank}(d_2)rank(d2)	Order-1 Residual (εharm\varepsilon_{\mathrm{harm}}εharm)	Order-2 Residual (εharm2\varepsilon^2_{\mathrm{harm}}εharm2: Cubes →\to→ All Z2Z_2Z2)	Structural Reading of the Tower
C8 ($K=3$)	6	4	16/16	2	2	0	$0 \to 0$	Closed at both route and relation orders.
Q3	3	3	44/49	1	2	0	$0 \to 0$	Cube

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Token Arena Substrate (KK Tokens)	**dim CG1\dim C^1_GdimCG1**	**dim CG2\dim C^2_GdimCG2**	**Commuting Cubes / dim Z2\dim Z_2dim Z2**	**rank (d1)\operatorname{rank}(d_1)rank(d1)**	**rank (d2)\operatorname{rank}(d_2)rank(d2)**	**Order-1 Residual (εharm\varepsilon_{\mathrm{harm}}εharm)**	**Order-2 Residual (εharm2\varepsilon^2_{\mathrm{harm}}εharm2: Cubes →\to→ All Z2Z_2Z2)**	**Structural Reading of the Tower**
($K = 3$)								strictly inside Z_2 ($44 < 49$); invariant sector clo

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Token Arena Substrate (KK Tokens)	**dim CG1\dim C^1_GdimCG1**	**dim CG2\dim C^2_GdimCG2**	**Commuting Cubes / dim Z2\dim Z_2dim Z2**	**rank (d1)\operatorname{rank}(d_1)rank(d1)**	**rank (d2)\operatorname{rank}(d_2)rank(d2)**	**Order-1 Residual (εharm\varepsilon_{\mathrm{harm}}εharm)**	**Order-2 Residual (εharm2\varepsilon^2_{\mathrm{harm}}εharm2: Cubes →\to→ All Z2Z_2Z2)**	**Structural Reading of the Tower**
								sed.
Petersen ($K=3$)	4	2	$145/155$	1	1	0	$0 \to 0$	Cubes strictly inside Z_2 ($145 < 155$); invarian

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Token Arena Substrate (KK Tokens)	**dim CG1\ dim C^1_GdimCG1**	**dim CG2\ dim C^2_GdimCG2**	**Commuting Cubes / dim Z2\dim Z_2dim Z2**	**rank (d1)\ operatorname{rank}(d_1)rank(d1)**	**rank (d2)\ operatorname{rank}(d_2)rank(d2)**	**Order-1 Residual (εharm\varepsilon_{\mathrm{harm}}εharm)**	**Order-2 Residual (εharm2\varepsilon^2_{\mathrm{harm}}εharm2: Cubes →\to→ All Z2Z_2Z2)**	**Structural Reading of the Tower**
								tsector closed.
prism ($K=3$)	3	3	4/6	1	1	0	$1 \to 0$	**Closed at route order ($\varepsilon_{\mathrm{harm}}$**

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Token Arena Substrate (KK Tokens)	dim CG1\dim C^1_GdimCG1	dim CG2\dim C^2_GdimCG2	Commuting Cubes / dim Z2\dim Z_2dim Z2	rank (d1)\operatorname{rank}(d_1)rank(d1)	rank (d2)\operatorname{rank}(d_2)rank(d2)	Order-1 Residual (εharm\varepsilon_{\mathrm{harm}}εharm)	Order-2 Residual (εharm2\varepsilon^2_{\mathrm{harm}}εharm2: Cubes →\to→ All Z2Z_2Z2)	Structural Reading of the Tower
								0), open one order up ($\varepsilon^2_{\mathrm{harm}}$ 1).
bull ($K=3$)	7	1	$0/0$	1	0	1	$0 \to 0$	Open at route

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Token Arena Substrate (KK Tokens)	dim CG1\dim C^1_GdimCG1	dim CG2\dim C^2_GdimCG2	Commuting Cubes / dim Z2\dim Z_2dim Z2	rank (d1)\operatorname{rank}(d_1)rank(d1)	rank (d2)\operatorname{rank}(d_2)rank(d2)	Order-1 Residual (εharm\varepsilon_{\mathrm{harm}}εharm)	Order-2 Residual (εharm2\varepsilon^2_{\mathrm{harm}}εharm2: Cubes →\to→ All Z2Z_2Z2)	Structural Reading of the Tower
								order ($\varepsilon_{\mathrm{harm}}1$), closed above ($\varepsilon^2_{\mathrm{harm}}0$).
caterpill	38	13	2/2	12	1	2	$0 \to 0$	Two surviv

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Token Arena Substrate (KK Tokens)	**dim CG1\ dim C^1_GdimCG1**	**dim CG2\ dim C^2_GdimCG2**	**Commuting Cubes / dim Z2\dim Z_2dim Z2**	**rank (d1)\ operatorname{rank}(d_1)rank(d1)**	**rank (d2)\ operatorname{rank}(d_2)rank(d2)**	**Order-1 Residual (εharm\varepsilon_{\mathrm{harm}}εharm)**	**Order-2 Residual (εharm2\varepsilon^2_{\mathrm{harm}}εharm2: Cubes →\to→ All Z2Z_2Z2)**	**Structural Reading of the Tower**
ar ($K=3$)								ing channels at route order, zero above.

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Token Arena Substrate (KK Tokens)	dim CG1\dim C^1_GdimCG1	dim CG2\dim C^2_GdimCG2	Commuting Cubes / dim Z2\dim Z_2dim Z2	rank (d1)\operatorname{rank}(d_1)rank(d1)	rank (d2)\operatorname{rank}(d_2)rank(d2)	Order-1 Residual (εharm\varepsilon_{\mathrm{harm}}εharm)	Order-2 Residual (εharm2\varepsilon^2_{\mathrm{harm}}εharm2: Cubes →\to→ All Z2Z_2Z2)	Structural Reading of the Tower
CE8 ($K=3$)	32	42	118/153	22	17	1	$3 \to 0$	Residuals of different rank at both orders; predicate-

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Token Arena Substrate (KK Tokens)	**dim CG1\ dim C^1_GdimCG1**	**dim CG2\ dim C^2_GdimCG2**	**Commuting Cubes / dim Z2\dim Z_2dim Z2**	**rank (d1)\ operatorname{rank}(d_1)rank(d1)**	**rank (d2)\ operatorname{rank}(d_2)rank(d2)**	**Order-1 Residual (εharm\varepsilon_{\mathrm{harm}}εharm)**	**Order-2 Residual (εharm2\varepsilon^2_{\mathrm{harm}}εharm2: Cubes →\to→ All Z2Z_2Z2)**	**Structural Reading of the Tower**
								dependent.
wheel 6 ($K = 3$)	6	3	5/5	3	0	0	$0 \to 0$	Route channels closed by $\mathcal{K}_{\mathrm{disj}}$ ($\varepsilon =$

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Token Arena Substrate (KK Tokens)	dim C G1\ dim C^1_ GdimC G1	dim C G2\ dim C^2_ GdimC G2	Commuting Cubes / dim Z2\dim Z_2dim Z2	rank (d1)\ operatorname{rank}(d_1)rank(d1)	rank (d2)\ operatorname{rank}(d_2)rank(d2)	Order-1 Residual (εharm\varepsilon_{\mathrm{harm}}εharm)	Order-2 Residual (εharm2\varepsilon^2_{\mathrm{harm}}εharm2: Cubes →\to→ All Z2Z_2Z2)	Structural Reading of the Tower
								$3 \to \varepsilon_{\mathrm{harm}}$ 0).

By Hodge duality, at every order of the complex, the surviving residual satisfies the universal descent identity (verified across $20/20$ order-1 cases and $12/12$ order-2 cases):

$$\varepsilon_{\mathrm{harm}} = \dim Z - \dim\mathrm{span}(\text{declared cells})$$

Here $\dim Z$ is the ceiling (the invariant cycle space at the empty predicate where nothing is declared equivalent), and $\dim\mathrm{span}(\text{declared cells})$ is the descent. This single equation explains why apparent zero readings divide into three distinct mathematical cases: genuine absence ($\varepsilon = 0$, as on the paw graph), route-order closure ($\varepsilon > 0, \varepsilon_{\mathrm{harm}} = 0$, as on W_6 where 3 channels are closed by 2-cells), and single-order screening ($\varepsilon_{\mathrm{harm}} = 0, \varepsilon^2_{\mathrm{harm}} > 0$, as on prism at $K = 3$, where a reader stopping at H^1 misses the surviving order-2 channel). Moreover, filtering the predicate poset from $\mathcal{K}_{\mathrm{disj}}$ to $\mathcal{K}_4$ by adding symmetry orbits one by one forms a persistence module: on two bridged 4-cycles ($K = 2, \varepsilon = 54$), the 7 harmonic channels die at 7 distinct orbit thresholds (+orbit1, +orbit3, +orbit4, +orbit8, +orbit10, +orbit19, +orbit20), whereas on CE8 ($K = 2, \varepsilon = 8$), the single harmonic channel survives all 9 extra orbits of size 4 and dies precisely at +orbit10, the first orbit of size 8.

Pressures, Flows, and the Dual-Wave Operational Machines: BBP and SHA-256

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When the compiler space is set in motion, its dynamics are governed by the interplay of stretch and fold, denominator pressure, value-free carry flows, and Landauer anti-aliasing constraints.

The BBP Machine: Compilation at Step Zero, Silent Carry Bands, and the Wheel Behind the Line

In *Nest* (Nest.docx), the Bailey–Borwein–Plouffe formula $\pi = 4L_1 - 2L_4 - L_5 - L_6$ with $L_j = \sum_{k=0}^{\infty} 16^{-k}/(8k+j)$ is decompiled from a sequential digit-extraction algorithm into a single simultaneous dynamical system. Defining the phase of lane (k, j) at step $n \geq 0$ by $x_{k,j}(n) = \{16^{n-k}/(8k+j)\}$, every lane—including the infinite "unborn" tail $k > n$—is compiled in full at step 0 ($S_0 = \sum_{k,j} c_j\, x_{k,j}(0) = \pi - 4$) and evolves under one universal law $x_{k,j}(n+1) = T(x_{k,j}(n)) = \{16x_{k,j}(n)\}$ (verified across $8{,}160$ exact rational phases and $8{,}004{,}000$ crossings over $2{,}000$ steps). A lane is not scheduled by an external controller; for $d = 8k + j \geq 2$, its phase nests (multiplying by 16 each step without crossing 1) until denominator pressure forces its first positive crossing $q_{k,j}(n) = \lfloor 16x_{k,j}(n) \rfloor$ at exact step $n = k + \lceil \log_{16} d \rceil - 1$ (verified on $7{,}988/7{,}988$ lanes). The $(n+1)$-th hexadecimal digit of π after the radix point is emitted as a difference of differences: $\delta_{n+1} = C_n + I_{n+1} - 16I_n$, where $C_n = \sum c_j q_{k,j}(n)$ and $I_n = \lfloor S_n \rfloor$.

When the four lanes of layer k are encapsulated into a single rational layer value $V_k = \sum_{j\in\{1,4,5,6\}} c_j/(8k+j) = (120k^2 + 151k + 47)/((8k+1)(2k+1)(8k+5)(4k+3))$, the zero-sum inversion in the weights ($\sum c_j = 4 - 2 - 1 - 1 = 0$ and $\sum j c_j = -15$) compresses the layer value by two orders of magnitude to $V_k = 15/(64k^2) + O(k^{-3})$. Consequently, while the first lane of layer k activates at age $m_1 = \lceil \log_{16}(8k+1) \rceil - 1$, the first nonzero hexadecimal value digit $w_k(m) = \lfloor 16\{16^m V_k\} \rfloor$ arrives later, at age $m_2 = \lceil \log_{16}(1/V_k) \rceil - 1 \approx \log_{16}(64k^2/15) - 1$. Between m_1 and m_2 lies a silent band (3 steps wide in 78% of layers and 2 steps wide in 18% of layers for $k \leq 10^4$) during which individual lanes fire actively while the layer's net value output is zero. Across all ages $m \geq 0$, the four-lane signed crossing signal $H_k(m) = \sum_j c_j\, q_{k,j}(k+m)$ and the true value digits $w_k(m)$ are related by a bounded internal carry state $J_k(m) = \lfloor \sum_j c_j\, x_{k,j}(k+m) \rfloor \in \{-4, \ldots, 3\}$ with $J_k(0) = 0$ (verified across $45{,}030$ ages):

$$H_k(m) = w_k(m) + 16J_k(m) - J_k(m+1), \qquad \sum_{m=0}^{\infty} H_k(m) 16^{-m-1} = V_k$$

Thus, the carry flow $16J_k(m) - J_k(m+1)$ dominates the local motion (variance 484 for H_k versus 21.4 for w_k) while contributing identically zero to the compiled value V_k.

Decompiling the BBP weights further exposes the complex wheel behind the real line. Let $s = \frac{1+i}{2} = \frac{1}{\sqrt{2}} e^{i\pi/4}$ be the quarter-turn scale-phase step. Multiplication by s on $\mathbb{R}^2$ has matrix $\begin{psmallmatrix} 1/2 & -1/2 \\ 1/2 & 1/2 \end{psmallmatrix}$ with area Jacobian $\det(M_s) = |s|^2 = 1/2$ (1 bit per tick), while its inverse $s^{-1} = 1 - i = 2\bar{s}$ acts on the Gaussian torus $\mathbb{C}/\mathbb{Z}[i]$ as an expanding endomorphism of degree $N(1-i) = 2$, index $|\mathbb{Z}[i]/(1-i)^k| = 2^k$, and topological entropy $\log 2 = 1$ bit/tick. Over 4 steps, $s^4 = -1/4$ (the half-turn seam that relates alternating Base -4 to Base 16 across two seams); over 8 steps, $s^8 = 1/16$ (and $s^{-8} = (1-i)^8 = 16$, unwinding the nest). Expanding $\sum_{n=1}^{\infty} s^n/n = -\log(1-s) = -\log\bar{s} = \frac{1}{2}\log 2 + i\frac{\pi}{4}$ separates the wheel into two orthogonal shadows under the mirror $s \mapsto \bar{s}$ (where $s + \bar{s} = 1$ and $s \perp \bar{s}$). The symmetric fold along the real axis produces $\log(1+i) + \log(1-i) = \log 2 = \sum_{j=1}^{8} 2\,\mathrm{Re}(s^j) L_j$, whose weights $(1, 0, -1/2, -1/2, -1/4, 0, 1/8, 1/8)$ automatically sum to zero because $\sum_{j=1}^{8} s^j = \frac{15}{16} i$ is purely

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imaginary. The antisymmetric inversion along the imaginary axis produces $\log(1+i)-\log(1-i)=i\pi/2 \Longrightarrow \pi=\sum_{j=1}^{8} 4\,\mathrm{Im}(s^j)L_j$ with wheel weights $c(-1/4)=(2,2,1,0,-1/2,-1/2,-1/4,0)$ summing to $15/4$; subtracting $1/4$ of the zero-route relation R shifts the wheel onto the crest $\sum c_j=0$ at $r=0$, yielding the sparse 4-lane BBP formula $(4,0,0,-2,-1,-1,0,0)$. Both constants are integrals of a single kernel $K(x)=\frac{2(1/\sqrt{2}-x)}{|1-e^{i\pi/4}x|^2}$ from 0 to 1: $\log 2=\int_0^1 K\,(x)\,dx$ and $\pi=\int_0^1 \frac{4K(x)}{1-x^2}\,dx$, where $K(x)$ changes sign at the exact crest $x=1/\sqrt{2}=|s|$.

Decompiling Partial-Inversion Lane Balance and the SHA-256 Constraint Field

The power of decompiling while computing is further demonstrated on two open frontier problems: the balance of partial-inversion lanes from *Nest* and the constraint-field inversion of SHA-256.

In *Nest*, roughly 4% of partial-inversion lanes (804 lanes tested) exhibited zero net lean, a phenomenon that produced two dead ends when measured from the outside as a scalar density. Decompiled in *One Program, Infinite Compilers*, every lane splits into a reader ring and a field ring: the lean is proven to be the reader's average of the field's discrepancy, holding identically across all 804 lanes.

In the SHA-256 architecture (nexus_compiler_space.py, Latest_published_papers.docx, and From-potential-to-matter…), the transformation grammar operates on an 8-word (256-bit) state $s=(a,b,c,d,e,f,g,h)$ driven by a 64-word expanded message schedule W_t and round constants K_t (the fractional cube roots of the first 64 primes). When the round drivers (W_t, K_t) are retained as observer context, the SHA-256 round transformation is an exact bijection (sha_round_inverse recovers the parent 256-bit state with 0 failures across $10{,}000$ random trials). Within each round $(a,b,c,d,e,f,g,h)\mapsto(t_1+t_2,a,b,c,d+t_1,e,f,g)$, 6 of the 8 register updates are pure wire relocations (generation-free transport moves), while only 2 words (a' and e') involve arithmetic mixing. The thermodynamic and informational crush occurs in only one place: the $512\to256$ width projection step, which acts as a coherent anti-aliasing filter rather than metaphysical annihilation.

Exhaustive functional-graph computation at reduced widths $w=16$ and $w=20$ proves that iterated hashing $H_{n+1}=\mathrm{SHA}(H_n)$ is a contractive funnel rather than a bijective wheel: image fractions drop as $0.6314N\to0.4684N\to0.3736N$ at width 16 and $0.6319N\to0.4681N\to0.3736N$ at width 20, matching random maps ($1-e^{-1}\approx0.63212$) to four decimal places, with successive step displacements orthogonal at $89.893^\circ\pm3.667^\circ$. Because image cardinality $|f(X)|$ is a representation-independent property of the map itself, the iterated-hash wheel hypothesis is dead in every representation. Similarly, the anomalously low carry-exhaust weight of the empty padded message (514 of $2{,}048$ vs. 768 ± 21 for random messages) is proven to be an artifact of seed Hamming-weight sparsity (weight 1 vs. random 129): a weight-matched sparse control (weight 3) yields 560, shares the exact zero-carry window across $t=16\ldots24$, and correlates with the empty trajectory at $+0.248$ after grammar centering.

Under constraint inversion, the terminal digest H is inserted into the fixed grammar as a boundary condition. At the Davies–Meyer feed-forward $H=S_{64}\boxplus IV\pmod{2^{32}}$, the global 256-bit constraint factors cleanly into 8 independent 32-bit word conditions $S_{64}[i]=(H[i]-IV[i])\bmod 2^{32}$, because modular addition couples carries strictly inside each 32-bit word and never across word boundaries. Walking backward from the digest, however, destroys this factorization in a single step: the ancestry of W_{63} encompasses all 16 seed

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words ($A(W_{63}) = \{0, \ldots, 15\}$), coupling all eight lanes globally after one inverse round. Walking forward from the seed reveals the opposite geometry: the first 10 expanded schedule words ($W_{16} \ldots W_{25}$) maintain narrow seed ancestry before reaching full 16-word coupling at W_{26}, while single-word seed variations leave an invariant free-cone seam at rounds $\{16,17\}$. Because a stateless function $\rho: (P, G) \to R$ cannot detect whether its output has entered a periodic relational orbit $Q_{n+r} \sim Q_n$ ($r \geq 2$, the only natively two-frame Co′-legal observable), any relational tuner must be a stateful transducer $\rho: (\Sigma, P, G) \to (\Sigma', R)$. Furthermore, because any deterministic loop consuming (H, F_{SHA}) defines a function $\kappa: H \to M'$, it can output at most one preimage per collision fiber; thus, a restricted domain $\mathcal{A}$ is recoverable if and only if $\mathcal{A}$ forms a partial section of the compression map, where linguistic or physical constraints supply the admission oversampling price $A(I) = -\log_2(|\mathrm{Adm}|/|S|)$ (collapsing a 16.00-member fiber to 1.73 with 52.5% unique preimages under two cheap constraints).

Synthesis and Forward Continuation Protocol

Taking the entire universe of transformations across the NEXUS corpus as one connected system demonstrates that rings, registers, bases, cohomological residuals, radix spigots, and atomic shells are not separate theories requiring forced unification; they are typed projections and recursive constructors over a single graded domain $S[b, n]$.

In $S[b, n]$, the displacement alphabet $\Delta[b, n]$ is the word layer $S[b, n]$ with uniform fibers of size b^n, and the operator space $T[b, L \to L]$ is the word layer $S[b, L \cdot b^L]$. In the 2-bit ring (00011011), this identity manifests directly: angular step shifts fold radially into wider window lengths via $w_i^{(L+1)} = 2w_i^{(L)} + s_{i+L}$, step gaps double along Frobenius squaring orbits modulo $2^8 - 1$, and at $L = 8$ the window width equals both the ring circumference and the 2-bit operator truth-table length, locking clockwise and counter-clockwise hexadecimal pairs into exact bit-reversal runtime reflection. Grouping symbols into a base is strictly the reader's step width (steps × width = LEN). Only power-of-two blocks ($b = 2,4,16,256$) act as local readings on the 64-bit register stack, whereas non-power-of-two bases ($b = 3,10,22$) act as global conversions. Across all prime-power lane machines $p^{n/2}$, binary ($p = 2$) is the unique base where scale returns ($\sqrt{p} \mp 1$) and phase returns ($\sqrt{p} - \zeta_n^m$ at $\mathrm{Re} w = 1/2$, the $.5$ potential midpoint) are simultaneously prime-free units, forcing $Z_2(2^e) = 1$ at every register layer via the Galois-stable core $R_0 = \bigcap_\sigma \pi_\sigma(R)$.

Embedding the nested rings into 3D space maps word length LEN to radial shell thickness, ordered route histories to the non-abelian interior, and symbol histograms to the abelianized spherical surface. Massless radiation sits at the single-route poles, while massive matter sits at the equatorial fold where spatial drift vanishes and internal turnaround ticks at $\omega = mc^2/\hbar$. Reading a part of size $\ell < n$ across a two-position gap yields a doubled 2ℓ-bit ring whose $2\ell + 1$ order-free surface latitudes carry exact Schwinger angular momentum $S^2 = \ell(\ell + 1)$, generating the shell capacities $2n^2$ and—via the inadmissibility ($n_r = -1$) of even-degree nodeless predecessors—forcing the paired periodic closures $2,10,18,36,54,86,118$ with zero free parameters.

To advance the programme along its sealed open boundaries without violating the non-conflation discipline established in *Two Quotients of One Machine* and *The Ceiling and the Descent*, three immediate computational continuations are specified. First, the all-n merge-to-interior proof and carry-seam bridge must formalize the inductive proof of the local Myhill–Nerode run-signature update U for the interior-merge

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case in the n-bit adder, constructing the explicit functor that maps the common carry-seam staircase structure onto both the continuation state count $F(S)$ and the disjoint-square cohomology rank $b_1^{\mathrm{disj}}(n) = \binom{n}{2} + (n-1)$. Second, the zero-cost base-scaling falsifier from the constraint frame should be executed by rendering the hexadecimal and decimal echo bands in Base 4 and Base 256 to verify whether fiber density and V-band splitting scale as $1/b$ (confirming bands as reader fibers $\rho^{-1}(r)$) or remain base-invariant (relocating bands to the primitive (P, G) substrate). Third, forward constraint propagation across the 10-word SHA-256 window ($W_{16} \ldots W_{25}$) and the $\{16,17\}$ free-cone seam should be evaluated using a stateful transducer reader ρ: $(\Sigma, P, G) \to (\Sigma', R)$ operating on fiber splits prior to scalar collapse, testing for incremental relational orbit locking ($Q_{n+r} \sim Q_n, r \geq 2$) against grammar-centered, weight-matched sparse controls.

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